Chapter 1. Fundamental principles of counting

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

1.1 Objects and boundary cases

Start with a finite set S. Specify whether objects are distinguishable, positions labeled, order significant, and repetition allowed, and which descriptions count as the same result. Every formula parameter encodes these choices.

Choose three objects from five distinguishable symbols.

Outcome	Order	Repetition	Count
Length-three string of distinct letters	Recorded	Forbidden	$5 \cdot 4 \cdot 3 = 60$
Length-three string	Recorded	Allowed	$5^3 = 125$
Three-element subset	Ignored	Forbidden	$\binom{5}{3} = 10$
Multiset of size three	Ignored	Allowed	$\binom{7}{3} = 35$

Use $0! = 1$. There is one empty sequence and one empty subset. Making no choices has one outcome; a set with no admissible outcome has size zero.

1.2 The sum rule is a disjoint partition

If $S = A_1 \sqcup \cdots \sqcup A_k$, then

$$|S| = \sum_i |A_i|.$$

Every object lies in exactly one class, so the right side counts it once. Overlaps require repartitioning or subtraction of duplicates; omitted classes cause undercounting.

For $A \subseteq U$, the partition $U = A \sqcup (U \setminus A)$ gives $|A| = |U| - |U \setminus A|$.

A four-digit PIN may begin with zero. Of 10^4 PINs, 9^4 have no 7, leaving $10^4 - 9^4 = 3439$ with at least one 7. For four-digit integers the first digit cannot be zero. Both the universe and complement must then be recounted.

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1.3 The product rule needs no probabilistic independence

Suppose every admissible prefix of length $i - 1$ has exactly n_i extensions. There are $\prod_{i=1}^{r} n_i$ outcomes of length r. Induction proves this by multiplying each level's node count by its common branching count.

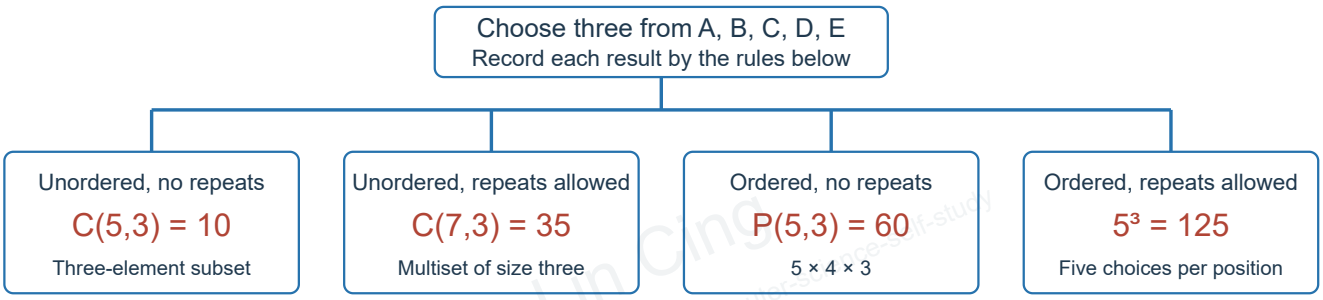
Available objects may depend on previous choices while their number stays fixed. Selecting distinct letters changes the second-step options, but always leaves four choices.

If first choosing A permits two continuations and choosing B permits three, the total is $2 + 3 = 5$. When branching counts differ, sum over prefix classes instead of multiplying an arbitrary count.

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1.3 The product rule needs no probabilistic independence (continued)

Five distinct letters: order and repetition define four different sample spaces



The models count strings, subsets, and multisets. Decide what data the result retains before choosing a formula.

1.4 Division requires equal fibers

Let $f : X \to Y$ be onto and give every $y \in Y$ exactly m preimages. Its disjoint fibers $f^{-1}(y)$ partition X, so

$$|X| = m|Y|, \qquad |Y| = |X|/m.$$

A division argument must say which labels are erased and how often each result was counted. Apparent symmetry does not prove constant multiplicity.

For three-position binary necklaces, identifying rotations gives singleton classes 000 and 111 and size-three classes represented by 001 and 011, four classes in total. The noninteger $2^3/3$ fails because fibers have unequal size. Chapter 16 uses Burnside's lemma for general symmetry counting; here division requires a proof of equal fibers.

1.5 Permutations and combinations share a forgetful map

For $n \geq 0, 0 \leq r \leq n$, ordered selection of distinct objects gives

$$P(n, r) = n(n-1)\cdots(n-r+1) = n!/(n-r)!.$$

Erasing order produces an r-subset, with exactly $r!$ permutations per subset. Thus

$$\binom{n}{r} = P(n, r)/r! = \frac{n!}{r!(n-r)!}.$$

For integer $n \geq 0$, use $\binom{n}{r} = 0$ if $r < 0$ or $r > n$. Do not arbitrarily put negative upper indices into the factorial formula.

When $r = 0$, both sides have only the empty object and multiplicity $0! = 1$. When $r > n$, selection without repetition is impossible.

1.6 Circular arrangements and reflection

Arrange $n \geq 1$ distinct objects around a circle, identifying rotations but distinguishing clockwise and counterclockwise order. Anchor a specified object and list the others clockwise. This bijection gives $(n-1)!$.

If reflection is also identified, each rotation class has a different reverse for $n \geq 3$. A reflection exchanges at least one pair of positions and cannot preserve all distinct labels. Exactly two classes merge, giving $(n-1)!/2$.

For $n = 1, 2$, only one circular arrangement exists, and reversal does not give two different rotation classes. Division by two fails. With repeated objects, one also cannot assume that every rotation gives n distinct images.

Five different people at a round table give $4! = 24$ directed circular orders, or 12 when only undirected adjacency matters and reflection is identified.

1.7 Multiset permutations

Let $n = n_1 + \cdots + n_k$, with n_i indistinguishable objects of type i. Temporarily label all copies to obtain $n!$ permutations. Erasing each type's internal labels gives exactly $\prod_i n_i!$ preimages per visible string. Therefore

$$\#\text{strings} = \frac{n!}{n_1! \cdots n_k!}.$$

The internal labels are an auxiliary construction proving equal fiber size.

TRANSMITTER has three T's, two R's, and one each of A, N, S, M, I, E, eleven letters total:

$$11!/(3!2!) = 3\,326\,400.$$

If the three T's are adjacent, contract them into a TTT block. Nine objects remain, with only R repeated, giving $9!/2! = 181\,440$. Contracting and expanding TTT are inverse maps, rather than an arbitrary reduction of the factorial.

1.8 Nonadjacency through gaps

BANANARAMA has five A's and five other letters B, N, N, R, M. Arrange the others in $5!/2! = 60$ ways, producing six gaps including both ends.

Nonadjacent A's occupy distinct gaps. Choose five of the six gaps:

$$60\binom{6}{5} = 360.$$

Deleting all A's uniquely recovers the other-letter arrangement and selected gaps, so no result is lost or duplicated.

For a fixed arrangement of m non-A objects, placing a indistinguishable nonadjacent A's gives $\binom{m+1}{a}$ possibilities, or zero if $a > m + 1$. Distinguishable A's require an additional $a!$ assignments to the selected gaps.

1.9 Compressing nonadjacent positions

Write r nonadjacent positions from $1, \dots, n$ as

$$1 \le x_1 < \cdots < x_r \le n, \quad x_{i+1} \ge x_i + 2.$$

Set $y_i = x_i - (i-1)$. Then

$$1 \le y_1 < \cdots < y_r \le n - r + 1.$$

The inverse $x_i = y_i + i - 1$ restores the restriction. For $r \ge 1, n \ge 2r - 1$, the count is $\binom{n-r+1}{r}$. For $n < 2r - 1$, it is zero; for $r = 0$, only the empty choice exists.

Three nonadjacent positions among eight give $\binom{6}{3} = 20$. Positions $(1, 4, 7)$ compress to $(1, 3, 5)$. The unrestricted $\binom{8}{3}$ includes forbidden adjacent choices.

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1.10 Symmetry, Pascal's identity, and marked subsets

Complementation bijects r-subsets and $(n-r)$-subsets, proving $\binom{n}{r} = \binom{n}{n-r}$.

For $n \ge 1$, classify by membership of a specified element:

$$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}.$$

The two classes are disjoint and exhaustive. The out-of-range convention covers $r = 0, n$.

Count an r-subset with a marked representative, choosing the subset first or representative first:

$$r\binom{n}{r} = n\binom{n-1}{r-1}.$$

The objects are pairs (S, s), $s \in S$. Both descriptions retain the extra role.

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1.11 Vandermonde and hockey-stick identities

Choose an r-subset from disjoint groups A, B of sizes a, b. Classify by its k members from A:

$$\sum_k \binom{a}{k}\binom{b}{r-k} = \binom{a+b}{r}.$$

Inadmissible terms are zero. With $a = 3, b = 4, r = 2$, the left side is $6 + 12 + 3 = 21 = \binom{7}{2}$.

Choose $r + 1$ elements of $\{1, \ldots, n + 1\}$, classified by their largest element $i + 1$. The remaining r come from the first i elements:

$$\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}.$$

The largest element uniquely determines the class, so terms cannot overlap. This is the hockey-stick identity for $0 \le r \le n$.

1.12 A coefficient proof of the binomial theorem

For nonnegative integer n, with commuting variables or in a commutative ring,

$$(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

Choose the k of the n factors supplying y; the others supply x. Choices are k-subsets, hence coefficient $\binom{n}{k}$.

Noncommuting objects need not allow combining cross terms. For matrices, $(A + B)^2 = A^2 + AB + BA + B^2$; the middle becomes $2AB$ only when $AB = BA$.

Setting $x = y = 1$ gives $\sum_k \binom{n}{k} = 2^n$. Setting $x = 1, y = -1$ gives equal numbers of even- and odd-size subsets for $n \ge 1$. Toggling a fixed element's membership is another bijection.

1.13 Multinomial coefficients and weighted terms

For integers $m \geq 1, n \geq 0, n_i \geq 0$ with $n_1 + \cdots + n_m = n$, assign the n_i factors supplying each x_i. The count is

$$\binom{n}{n_1, \ldots, n_m} = \frac{n!}{\prod_i n_i!}.$$

Therefore

$$(x_1 + \cdots + x_m)^n = \sum_{n_1 + \cdots + n_m = n} \frac{n!}{\prod_i n_i!} \prod_i x_i^{n_i}.$$

Numerical coefficients must also be multiplied. For example,

$$[x^3 y^2 z](2x - y + 3z)^6 = \frac{6!}{3!2!1!}\, 2^3 (-1)^2\, 3 = 1440.$$

A requested total degree other than six has coefficient zero. The homogeneous-degree constraint must hold before the factorial formula applies.

1.14 Weighted sums by marking roles

The sum $\sum_k k\binom{n}{k}$ counts a nonempty subset with a representative. Choose the representative in n ways and let each other person enter or stay out. For $n \geq 1$,

$$\sum_k k\binom{n}{k} = n2^{n-1}.$$

Counting subsets with an ordered pair of distinct representatives gives, for $n \geq 2$,

$$\sum_k k(k-1)\binom{n}{k} = n(n-1)2^{n-2}.$$

As $k^2 = k(k-1) + k$,

$$\sum_k k^2\binom{n}{k} = n(n-1)2^{n-2} + n2^{n-1}.$$

For $n = 0, 1$, inspect the original sum. A form such as $0 \cdot 2^{-1}$ should not hide the counting assumptions.

1.15 Stars and bars as a decodable string

For $n \geq 1, r \geq 0$, nonnegative solutions of $x_1 + \cdots + x_n = r$ correspond to strings of r stars and $n-1$ bars. Segment i has x_i stars; adjacent bars represent a zero entry.

For example, $(2, 0, 3, 1)$ encodes as

$$** \,|\,| \, *** \,|\, *.$$

Counting stars in each segment uniquely inverts the map. Choose $n-1$ bar positions from $r+n-1$:

$$\#\{x_i \geq 0,\ \sum x_i = r\} = \binom{r+n-1}{n-1}.$$

For $r = 0$, only the all-zero solution exists; for $n = 1$, only $x_1 = r$. For zero variables, the empty sum is zero, giving one solution for total zero and none otherwise. Handle this separately instead of using a negative upper index.

1.16 Lower bounds, upper bounds, and inequalities

For $x_i \geq a_i$, put $y_i = x_i - a_i \geq 0$, leaving total $R = r - \sum_i a_i$. There are no solutions if $R < 0$; otherwise there are $\binom{R+n-1}{n-1}$. In particular, entries at least one give $\binom{r-1}{n-1}$ when $r \geq n$.

For $\sum_{i=1}^{n} x_i \leq r$, add slack $x_{n+1} = r - \sum_i x_i \geq 0$. This bijection to $n+1$ variables totaling r gives $\binom{r+n}{n}$.

One upper bound can use complements. For $x_1 + x_2 + x_3 = 7,\ 0 \leq x_1 \leq 2$, all solutions number $\binom{9}{2} = 36$. Inadmissible solutions have $x_1 \geq 3$. Substituting $y_1 = x_1 - 3$ leaves total four and $\binom{6}{2} = 15$ solutions. The answer is $36 - 15 = 21$. Multiple violations can overlap and require Chapter 8's inclusion-exclusion.

1.17 Labeled shelves and internal order

Place m distinct books on $n \geq 1$ labeled shelves, ordered left to right on each, with none empty. Permute all books in $m!$ ways and choose $n-1$ cuts among $m-1$ internal gaps:

$$m!\binom{m-1}{n-1}, \qquad m \geq n.$$

Concatenating shelves in label order uniquely recovers the permutation and cuts. For $m < n$, no placement exists.

If empty shelves are allowed, choose a length vector $x_1 + \cdots + x_n = m, x_i \geq 0$, then cut a book permutation to those lengths:

$$m!\binom{m+n-1}{n-1}.$$

Merely assigning each book to a shelf, without recording internal order, gives n^m. These models retain different data despite both being book-distribution problems.

1.18 Runs and positive compositions

A run is a maximal consecutive block of one symbol. For example,

$$110001101 = 11\,|\,000\,|\,11\,|\,0\,|\,1$$

has five runs, three of 1's and two of 0's. Suppose there are $a \geq 1$ ones and $b \geq 1$ zeros.

- For $2k$ runs, either symbol starts, with k runs of each. The count is $2\binom{a-1}{k-1}\binom{b-1}{k-1}$.
- For $2k+1$ runs, initial and final symbols agree. The count is

$$\binom{a-1}{k}\binom{b-1}{k-1} + \binom{a-1}{k-1}\binom{b-1}{k}.$$

Run lengths are positive. Their compositions and the starting symbol uniquely determine the word.

For $a = 5, b = 4$, exactly five runs give $6 \cdot 3 + 4 \cdot 3 = 30$. A nonempty single-symbol word has one run; the empty word has zero. Neither satisfies the two-positive-count assumption.

1.19 Exact counts in nested loops

Each innermost iteration increments the counter once.

```
counter = 0
for i = 1,...,n
    for j = 1,...,i
        for k = 1,...,j
            counter += 1
```

The objects are $1 \le k \le j \le i \le n$, rather than three independent choices of size n. The bijection

$$(k, j, i) \longmapsto (k, j + 1, i + 2)$$

produces three-element subsets of $\{1, \dots, n + 2\}$, giving $\binom{n+2}{3}$.

For $n = 8$, successive outer-loop counts are $1, 3, 6, 10, 15, 21, 28, 36$, totaling 120, equal to $\binom{10}{3}$. For $n = 0$, the empty loop gives zero.

1.20 Three distribution models

Fix m objects and $n \ge 1$ labeled boxes.

Object identities and recorded data	Empty boxes	Count
Distinguishable objects, destination only	Allowed	n^m
Indistinguishable objects, box counts only	Allowed	$\binom{m+n-1}{n-1}$
Distinguishable objects, internal linear order too	Allowed	$m!\binom{m+n-1}{n-1}$

If boxes are also indistinguishable, their labels are erased. Empty boxes and equal-size boxes cause unequal fibers, so division by $n!$ is not generally valid. Distinguishable objects partitioned into unlabeled nonempty blocks lead to Chapter 5's Stirling numbers. Unlabeled groups of indistinguishable objects lead to Chapter 9's integer partitions.

Keep the models distinct; the latter tools are not synonyms for stars and bars.

1.21 Catalan objects through admissible prefixes

For n pairs of parentheses, encode opening as $+1$, closing as -1, and let h_t be prefix height. Admissibility requires

$$h_0 = h_{2n} = 0, \qquad h_t \geq 0 \text{ for every } t.$$

Total height zero is insufficient. The word `())(()` has heights $1, 0, -1, 0, 1, 0$ and fails at step three.

There are $\binom{2n}{n}$ words with equal symbol counts. Subtract exactly those ever becoming negative. Equal counts do not justify declaring half admissible.

For $n = 0$, the empty word is admissible, so $C_0 = 1$. The reflection proof first takes $n \geq 1$.

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1.22 Reflection and its inverse

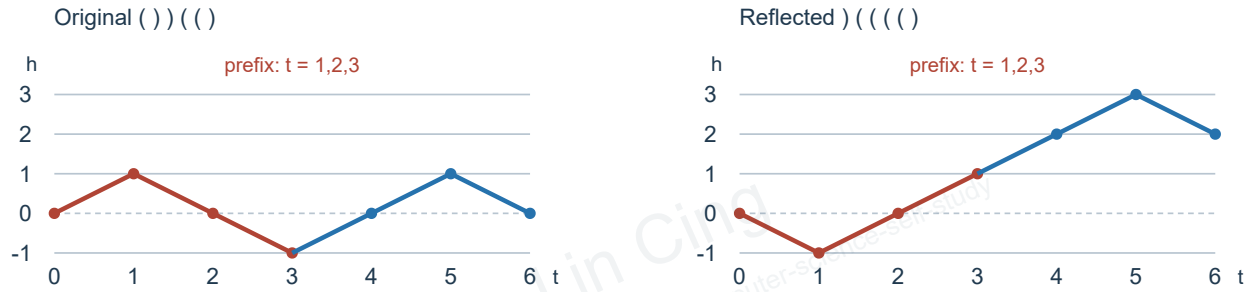
For an inadmissible balanced word, take the first time t with $h_t = -1$. Swap opening and closing symbols in positions $1, \ldots, t$. That prefix had one extra closing symbol, so the full transformed word has $n + 1$ openings, $n - 1$ closings, and endpoint height two.

Conversely, any word with these counts must first reach $+1$, since it ends at two. Swap its prefix through that time. The result is balanced and first reaches -1 there. These operations are inverse, hence a bijection.

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1.22 Reflection and its inverse (continued)

Catalan reflection: swap only the prefix through the first height −1



Left: first h=−1 at t=3. Right: first h=+1 at t=3. Swapping that prefix again recovers the original.

On the left, `())(()` first becomes negative at $t = 3$. Reflection gives `)((((`, whose first $+1$ also occurs at $t = 3$. Only the specified prefix is reflected.

1.23 The Catalan formula and values

The reflection target contains $\binom{2n}{n-1}$ words, so

$$\begin{aligned} C_n &= \binom{2n}{n} - \binom{2n}{n-1} \\ &= \left(1 - \frac{n}{n+1}\right)\binom{2n}{n} = \frac{1}{n+1}\binom{2n}{n}. \end{aligned}$$

The first terms are $1, 1, 2, 5, 14, 42, 132$.

For $n = 3$, the admissible words are

$$((())), \quad (()()), \quad (())(), \quad ()(()), \quad ()()().$$

Of $\binom{6}{3} = 20$ balanced words, $\binom{6}{2} = 15$ are inadmissible, leaving five. For $n = 6$, the count is $924 - 792 = 132$, rather than $924/2$.

1.24 Unique decomposition at the first return

A nonempty admissible word starts with an opening parenthesis. Its matching closing parenthesis is exactly the first return to height zero. Thus the word has the unique form

$$(U)V,$$

where U, V are admissible. If U has i pairs, V has $n-1-i$. Conversely the construction remains admissible, giving

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}, \qquad C_0 = 1.$$

For $n = 3$, $C_3 = C_0C_2 + C_1C_1 + C_2C_0 = 2 + 1 + 2 = 5$.

Dynamic programming computes the first n terms in $O(n^2)$ integer additions and multiplications and $O(n)$ stored integers. This is an arithmetic-operation count; growing-integer multiplication does not have constant bit cost.

1.25 Ordered full binary trees and parenthesizations

Every internal node of an ordered full binary tree has exactly a left and right child; leaves have none. With n internal nodes there are $n+1$ leaves. A single leaf proves the base $n = 0$. Joining subtrees with i, j internal nodes gives $n = i + j + 1$, and induction gives $(i+1) + (j+1) = n + 1$ leaves.

Encode a leaf by the empty word. If left and right subtrees encode as U, V, encode the parent as $(U)V$. First-return decomposition uniquely decodes both subtrees. Induction bijects trees with n internal nodes and admissible words with n pairs.

Place $n+1$ factors in fixed order at the leaves, with internal nodes representing binary multiplication. This gives C_n full parenthesizations, without permuting factors:

$$((ab)c)d,\ (a(bc))d,\ (ab)(cd),\ a((bc)d),\ a(b(cd)).$$

These are the five parenthesizations of four factors. Even if multiplication is associative, the evaluation structures differ.

1.26 Triangulations of a convex polygon

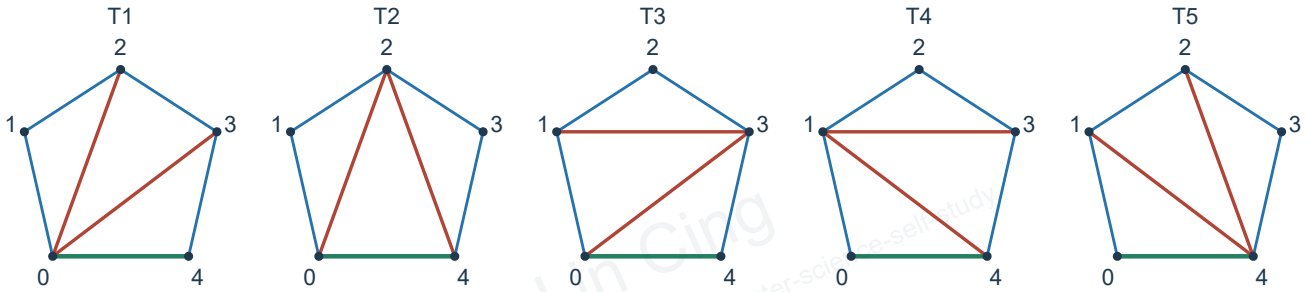
Fix a labeled convex $(n+2)$-gon, $n \geq 1$, with root edge 0–$(n+1)$. Each triangulation has a unique triangle on that edge. Its third vertex $i+1$ splits the rest into two smaller rooted polygons with counts C_i, C_{n-1-i}.

Conversely, the third vertex and two smaller triangulations uniquely reconstruct the original. The same Catalan recurrence results. A degenerate one-edge subproblem has count one, corresponding to C_0; a triangle has one triangulation. Equal initial values and recurrence give C_n.

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1.26 Triangulations of a convex polygon (continued)

Five triangulations of one labeled pentagon, each with two noncrossing diagonals



Green root edge 0–4. Its unique triangle splits the polygon into two smaller rooted polygons.

A pentagon has $C_3 = 5$ triangulations. Rotated labeled positions remain different; there is no further division by five.

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1.27 Reflection with unequal endpoint counts

Let $a \geq b \geq 0$. There are $\binom{a+b}{b}$ words with a steps $+1$ and b steps -1. Reflect an inadmissible word through its first height -1. The target has $a+1$ positive and $b-1$ negative steps.

Its endpoint height is $a-b+2 \geq 2$, so the first visit to $+1$ provides the same inverse. If $b=0$, no word is inadmissible and the second term is zero by convention. Hence

$$\binom{a+b}{b} - \binom{a+b}{b-1} = \frac{a-b+1}{a+1}\binom{a+b}{b}.$$

For $a=6, b=4$, this gives $210-120=90$. Setting $a=b=n$ recovers Catalan numbers. For $a<b$, the endpoint is already negative and the answer is zero. The formula's assumptions then fail.

1.28 Two counts of the same team

From six professors and nine doctoral students, select two professors and three students, then a leader:

$$\binom{6}{2}\binom{9}{3} \cdot 5 = 15 \cdot 84 \cdot 5 = 6300.$$

Alternatively select the leader first and classify by role:

$$\text{Professor leader: } 6\binom{5}{1}\binom{9}{3} = 2520,$$

$$\text{Student leader: } 9\binom{6}{2}\binom{8}{2} = 3780.$$

The disjoint exhaustive classes again total 6300. The three students have no ordered offices, so no extra $3!$ appears. The leader is an explicit extra mark that must be retained.

Both counts describe a team together with its leader; a team alone is a different object.

1.29 A formula needs an encoding and inverse

The chapter's checkable relationships use disjoint classes, prefix extensions, equal fibers, forgetting order, gap compression, stars and bars, run lengths, Catalan first returns, and reflection.

For a new problem, give the value and explain how every admissible object is encoded, why encoding is unique, how to recover the object, and how empty and minimal cases behave. Finite enumeration can expose omissions or duplicates. General formulas still require bijections, partitions, or double-counting proofs.

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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